

# Supplementary material for

“Leakage-controlled stacking and recursive feature elimination for type 2 diabetes prediction: a rigorous evaluation on the PIMA dataset”

Subrahmanyam Tanala

## S1 Validation design: where each loop runs

Figure S1 shows the complete validation design. There are three nested levels. The *outer* level is the 5×10-fold repeated cross-validation used for model comparison. Within each outer training part there are two *inner* levels: the 5-fold cross-validation that RFECV uses to choose the subset size, and the 5-fold split that produces out-of-fold probabilities for the stacking meta-learner. The 154-patient hold-out set is separated before any modelling and is used once, by the final pipeline refitted on all 614 development patients.

## S2 Statistical testing procedure

**Repeated cross-validation** Let  $x_{rk}$  be the value of a metric on fold  $k$  of repeat  $r$  ( $r = 1, \dots, 5$ ;  $k = 1, \dots, 10$ ;  $J = 50$ ). The point estimate is  $\bar{x} = J^{-1} \sum_{r,k} x_{rk}$ . Fold estimates share most of their training data, so the naive variance  $s^2/J$  is too small. Following Nadeau and Bengio, we use

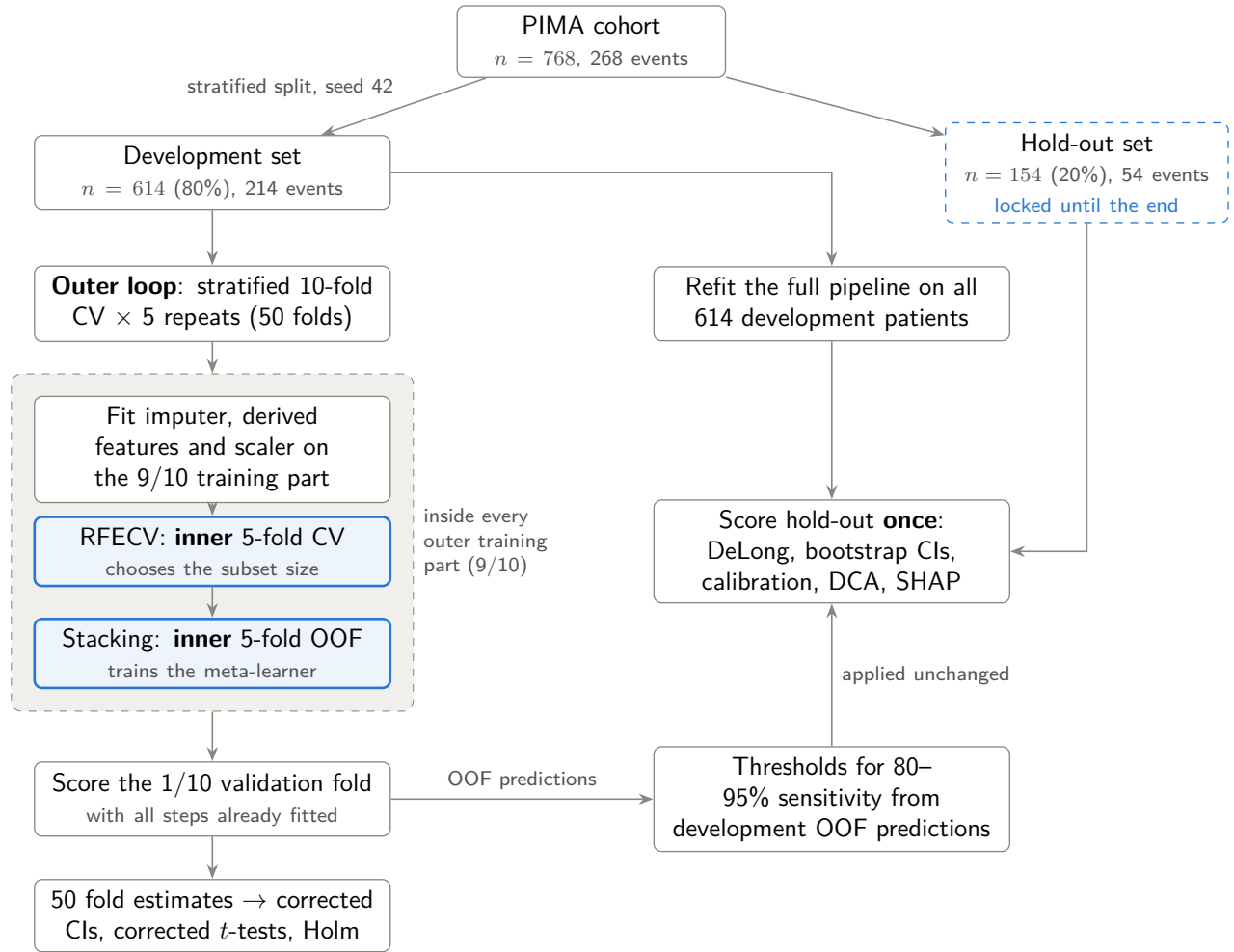
$$\widehat{\text{Var}}(\bar{x}) = \left( \frac{1}{J} + \frac{n_{\text{test}}}{n_{\text{train}}} \right) s^2, \quad \frac{n_{\text{test}}}{n_{\text{train}}} = \frac{1}{9}, \quad (\text{S1})$$

with  $s^2$  the sample variance of the  $J$  fold estimates. 95% confidence intervals are  $\bar{x} \pm t_{0.975, J-1} \sqrt{\widehat{\text{Var}}}$ .

**Paired comparisons** For a configuration  $A$  and the reference  $B$  (logistic regression on all features),  $d_{rk} = x_{rk}^A - x_{rk}^B$  is computed on identical folds. The corrected resampled statistic is  $t = \bar{d} / \sqrt{(1/J + 1/9) s_d^2}$ , referred to a  $t$  distribution with  $J - 1 = 49$  degrees of freedom (two-sided). The 22  $p$ -values (11 learners × 2 feature sets, plus LR-ranked RFECV, minus the reference) are adjusted with the Holm step-down procedure.

**Why this test** The corrected resampled  $t$ -test is designed for exactly this setting, where fold estimates from repeated  $k$ -fold cross-validation are correlated. In simulation studies it keeps the type I error close to the nominal level and replicates well across random partitions, whereas the uncorrected paired  $t$ -test is anti-conservative. Alternatives such as a nested bootstrap of the whole pipeline or Bayesian correlated  $t$ -tests are valid. However, a nested bootstrap would have required thousands of refits of RFECV and the stack (about 45 s each), and the conclusions here are far from any decision boundary: the stacking–logistic-regression difference is +0.001 with an adjusted  $p = 1.00$ .

**Hold-out set** ROC-AUC confidence intervals and pairwise comparisons on the 154 hold-out patients use DeLong’s non-parametric covariance (fast mid-rank implementation), with Holm adjustment over six comparisons against stacking + RFECV. All other metrics use 95% percentile bootstrap intervals from 2000 resamples of hold-out patients; resamples containing a single class are discarded.



**Figure S1:** Nested validation design. Blue boxes mark the two inner resampling loops that run inside every outer training part. Nothing from an outer validation fold or from the hold-out set is used to fit any step.

**Calibration and thresholds** The calibration intercept is estimated from  $\text{logit } P(y = 1) = a + \text{logit } \hat{p}$  (offset model) and the slope from  $\text{logit } P(y = 1) = a' + b \text{logit } \hat{p}$ , both by Newton–Raphson maximum likelihood. For a target sensitivity  $s$ , the threshold is the largest value  $t$  for which at least  $\lceil s n_+ \rceil$  of the  $n_+$  development-set positives have an out-of-fold probability (averaged over the five repeats) of at least  $t$ .

### S3 Full cross-validated results

**Table S1:** All 23 configurations from 5×10-fold repeated cross-validation on the development set ( $n = 614$ ). Means with Nadeau–Bengio corrected 95% CIs beneath. “+ RFECV” is the common random-forest-ranked RFECV subset. The last two columns give the unadjusted and Holm-adjusted corrected resampled  $t$ -test  $p$ -values for the ROC-AUC difference from logistic regression on all features. Threshold-based metrics use 0.5.

| Model                       | Feat. | Acc.                          | Bal. acc.                     | Prec.                         | Sens.                         | Spec.                         | F1                            | MCC                           | ROC-AUC                       | Brier                         | $p$    | $p_{\text{Holm}}$ |
|-----------------------------|-------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|--------|-------------------|
| Stacking                    | 16.0  | <b>0.759</b><br>(0.719–0.799) | <b>0.760</b><br>(0.722–0.797) | <b>0.635</b><br>(0.579–0.691) | <b>0.763</b><br>(0.710–0.815) | <b>0.757</b><br>(0.700–0.814) | <b>0.689</b><br>(0.646–0.732) | <b>0.505</b><br>(0.431–0.580) | <b>0.846</b><br>(0.812–0.880) | <b>0.164</b><br>(0.144–0.183) | 0.586  | 1.000             |
| Stacking + RFECV            | 12.0  | <b>0.756</b><br>(0.714–0.797) | <b>0.755</b><br>(0.715–0.795) | <b>0.631</b><br>(0.574–0.688) | <b>0.753</b><br>(0.697–0.810) | <b>0.757</b><br>(0.700–0.814) | <b>0.684</b><br>(0.638–0.730) | <b>0.497</b><br>(0.418–0.575) | <b>0.843</b><br>(0.809–0.878) | <b>0.165</b><br>(0.144–0.185) | 0.890  | 1.000             |
| Logistic regression         | 16.0  | <b>0.764</b><br>(0.727–0.801) | <b>0.759</b><br>(0.725–0.794) | <b>0.646</b><br>(0.592–0.700) | <b>0.744</b><br>(0.690–0.798) | <b>0.775</b><br>(0.721–0.828) | <b>0.688</b><br>(0.647–0.729) | <b>0.507</b><br>(0.438–0.576) | <b>0.842</b><br>(0.809–0.875) | <b>0.163</b><br>(0.144–0.183) | –      | –                 |
| Logistic regression + RFECV | 12.0  | <b>0.760</b><br>(0.721–0.798) | <b>0.750</b><br>(0.713–0.786) | <b>0.647</b><br>(0.588–0.706) | <b>0.717</b><br>(0.664–0.770) | <b>0.782</b><br>(0.727–0.838) | <b>0.676</b><br>(0.633–0.720) | <b>0.492</b><br>(0.418–0.566) | <b>0.842</b><br>(0.811–0.873) | <b>0.163</b><br>(0.145–0.182) | 0.997  | 1.000             |
| LR (LR-ranked RFECV)        | 10.1  | <b>0.760</b><br>(0.725–0.796) | <b>0.754</b><br>(0.717–0.790) | <b>0.641</b><br>(0.591–0.692) | <b>0.731</b><br>(0.671–0.791) | <b>0.776</b><br>(0.728–0.824) | <b>0.680</b><br>(0.637–0.724) | <b>0.496</b><br>(0.426–0.567) | <b>0.840</b><br>(0.808–0.873) | <b>0.164</b><br>(0.145–0.183) | 0.487  | 1.000             |
| SVM                         | 16.0  | <b>0.775</b><br>(0.735–0.815) | <b>0.747</b><br>(0.705–0.790) | <b>0.695</b><br>(0.624–0.767) | <b>0.655</b><br>(0.587–0.723) | <b>0.840</b><br>(0.790–0.889) | <b>0.670</b><br>(0.614–0.726) | <b>0.505</b><br>(0.418–0.591) | <b>0.840</b><br>(0.802–0.878) | <b>0.156</b><br>(0.138–0.175) | 0.796  | 1.000             |
| SVM + RFECV                 | 12.0  | <b>0.765</b><br>(0.722–0.808) | <b>0.736</b><br>(0.693–0.780) | <b>0.680</b><br>(0.605–0.755) | <b>0.643</b><br>(0.578–0.708) | <b>0.830</b><br>(0.776–0.884) | <b>0.656</b><br>(0.599–0.713) | <b>0.483</b><br>(0.393–0.572) | <b>0.837</b><br>(0.800–0.873) | <b>0.158</b><br>(0.140–0.176) | 0.577  | 1.000             |
| Extra trees + RFECV         | 12.0  | <b>0.753</b><br>(0.713–0.794) | <b>0.739</b><br>(0.698–0.780) | <b>0.642</b><br>(0.581–0.703) | <b>0.693</b><br>(0.627–0.759) | <b>0.786</b><br>(0.732–0.840) | <b>0.662</b><br>(0.612–0.712) | <b>0.474</b><br>(0.393–0.556) | <b>0.835</b><br>(0.798–0.872) | <b>0.161</b><br>(0.142–0.179) | 0.429  | 1.000             |
| Extra trees                 | 16.0  | <b>0.753</b><br>(0.707–0.798) | <b>0.735</b><br>(0.690–0.780) | <b>0.647</b><br>(0.576–0.717) | <b>0.676</b><br>(0.609–0.743) | <b>0.794</b><br>(0.736–0.852) | <b>0.656</b><br>(0.601–0.712) | <b>0.469</b><br>(0.377–0.560) | <b>0.835</b><br>(0.795–0.874) | <b>0.160</b><br>(0.140–0.181) | 0.448  | 1.000             |
| Naive Bayes + RFECV         | 12.0  | <b>0.762</b><br>(0.725–0.799) | <b>0.730</b><br>(0.692–0.768) | <b>0.682</b><br>(0.607–0.756) | <b>0.624</b><br>(0.564–0.683) | <b>0.836</b><br>(0.787–0.886) | <b>0.646</b><br>(0.595–0.698) | <b>0.473</b><br>(0.393–0.554) | <b>0.832</b><br>(0.797–0.867) | <b>0.205</b><br>(0.174–0.235) | 0.263  | 1.000             |
| Random forest               | 16.0  | <b>0.756</b><br>(0.710–0.802) | <b>0.746</b><br>(0.698–0.793) | <b>0.641</b><br>(0.575–0.707) | <b>0.712</b><br>(0.639–0.785) | <b>0.780</b><br>(0.724–0.836) | <b>0.671</b><br>(0.613–0.728) | <b>0.484</b><br>(0.390–0.578) | <b>0.831</b><br>(0.795–0.868) | <b>0.164</b><br>(0.144–0.184) | 0.269  | 1.000             |
| Random forest + RFECV       | 12.0  | <b>0.759</b><br>(0.715–0.804) | <b>0.750</b><br>(0.705–0.794) | <b>0.645</b><br>(0.580–0.711) | <b>0.718</b><br>(0.652–0.785) | <b>0.781</b><br>(0.725–0.838) | <b>0.676</b><br>(0.622–0.730) | <b>0.492</b><br>(0.403–0.581) | <b>0.831</b><br>(0.797–0.866) | <b>0.164</b><br>(0.145–0.184) | 0.262  | 1.000             |
| Gradient boosting           | 16.0  | <b>0.746</b><br>(0.709–0.784) | <b>0.711</b><br>(0.670–0.752) | <b>0.657</b><br>(0.589–0.726) | <b>0.595</b><br>(0.520–0.669) | <b>0.828</b><br>(0.779–0.876) | <b>0.618</b><br>(0.562–0.675) | <b>0.436</b><br>(0.353–0.518) | <b>0.831</b><br>(0.798–0.863) | <b>0.164</b><br>(0.145–0.183) | 0.294  | 1.000             |
| Naive Bayes                 | 16.0  | <b>0.773</b><br>(0.739–0.807) | <b>0.739</b><br>(0.703–0.775) | <b>0.701</b><br>(0.636–0.766) | <b>0.626</b><br>(0.566–0.686) | <b>0.852</b><br>(0.808–0.895) | <b>0.657</b><br>(0.608–0.706) | <b>0.494</b><br>(0.420–0.568) | <b>0.830</b><br>(0.793–0.867) | <b>0.205</b><br>(0.175–0.235) | 0.159  | 1.000             |
| Gradient boosting + RFECV   | 12.0  | <b>0.748</b><br>(0.709–0.788) | <b>0.714</b><br>(0.671–0.757) | <b>0.660</b><br>(0.588–0.733) | <b>0.601</b><br>(0.525–0.677) | <b>0.827</b><br>(0.776–0.878) | <b>0.623</b><br>(0.564–0.682) | <b>0.441</b><br>(0.354–0.528) | <b>0.830</b><br>(0.798–0.862) | <b>0.164</b><br>(0.145–0.183) | 0.291  | 1.000             |
| KNN                         | 16.0  | <b>0.760</b><br>(0.718–0.802) | <b>0.726</b><br>(0.681–0.770) | <b>0.681</b><br>(0.605–0.757) | <b>0.611</b><br>(0.541–0.680) | <b>0.841</b><br>(0.790–0.891) | <b>0.639</b><br>(0.578–0.700) | <b>0.466</b><br>(0.375–0.558) | <b>0.824</b><br>(0.781–0.866) | <b>0.162</b><br>(0.141–0.183) | 0.108  | 1.000             |
| XGBoost                     | 16.0  | <b>0.744</b><br>(0.701–0.787) | <b>0.713</b><br>(0.670–0.757) | <b>0.650</b><br>(0.576–0.723) | <b>0.611</b><br>(0.543–0.678) | <b>0.816</b><br>(0.761–0.871) | <b>0.625</b><br>(0.567–0.683) | <b>0.436</b><br>(0.346–0.527) | <b>0.821</b><br>(0.784–0.857) | <b>0.173</b><br>(0.149–0.196) | 0.056  | 0.954             |
| KNN + RFECV                 | 12.0  | <b>0.757</b><br>(0.713–0.800) | <b>0.721</b><br>(0.674–0.769) | <b>0.675</b><br>(0.600–0.750) | <b>0.604</b><br>(0.525–0.683) | <b>0.839</b><br>(0.788–0.889) | <b>0.632</b><br>(0.567–0.697) | <b>0.458</b><br>(0.361–0.554) | <b>0.821</b><br>(0.777–0.865) | <b>0.163</b><br>(0.143–0.183) | 0.110  | 1.000             |
| XGBoost + RFECV             | 12.0  | <b>0.746</b><br>(0.707–0.785) | <b>0.716</b><br>(0.674–0.757) | <b>0.651</b><br>(0.583–0.719) | <b>0.615</b><br>(0.543–0.687) | <b>0.817</b><br>(0.764–0.869) | <b>0.627</b><br>(0.572–0.682) | <b>0.440</b><br>(0.357–0.524) | <b>0.820</b><br>(0.785–0.854) | <b>0.172</b><br>(0.150–0.193) | 0.033  | 0.586             |
| LightGBM                    | 16.0  | <b>0.731</b><br>(0.687–0.775) | <b>0.706</b><br>(0.660–0.752) | <b>0.620</b><br>(0.551–0.690) | <b>0.622</b><br>(0.546–0.698) | <b>0.789</b><br>(0.733–0.846) | <b>0.617</b><br>(0.556–0.677) | <b>0.414</b><br>(0.321–0.508) | <b>0.803</b><br>(0.764–0.842) | <b>0.205</b><br>(0.172–0.237) | 0.002  | 0.044             |
| LightGBM + RFECV            | 12.0  | <b>0.733</b><br>(0.691–0.775) | <b>0.709</b><br>(0.667–0.750) | <b>0.625</b><br>(0.559–0.691) | <b>0.627</b><br>(0.559–0.695) | <b>0.790</b><br>(0.733–0.848) | <b>0.621</b><br>(0.568–0.674) | <b>0.420</b><br>(0.336–0.505) | <b>0.800</b><br>(0.765–0.836) | <b>0.202</b><br>(0.175–0.230) | <0.001 | 0.020             |
| Decision tree + RFECV       | 12.0  | <b>0.710</b><br>(0.668–0.752) | <b>0.723</b><br>(0.681–0.765) | <b>0.565</b><br>(0.517–0.613) | <b>0.767</b><br>(0.702–0.832) | <b>0.679</b><br>(0.626–0.733) | <b>0.648</b><br>(0.601–0.696) | <b>0.429</b><br>(0.348–0.510) | <b>0.772</b><br>(0.731–0.813) | <b>0.207</b><br>(0.180–0.235) | <0.001 | 0.020             |
| Decision tree               | 16.0  | <b>0.710</b><br>(0.669–0.751) | <b>0.725</b><br>(0.683–0.766) | <b>0.564</b><br>(0.518–0.610) | <b>0.774</b><br>(0.708–0.839) | <b>0.675</b><br>(0.622–0.729) | <b>0.650</b><br>(0.604–0.696) | <b>0.431</b><br>(0.352–0.511) | <b>0.770</b><br>(0.729–0.811) | <b>0.210</b><br>(0.181–0.239) | 0.001  | 0.025             |

## S4 Full hold-out results

**Table S2:** All hold-out metrics ( $n = 154$ , 54 events) at a 0.5 threshold. 95% CIs beneath: DeLong for ROC-AUC, percentile bootstrap (2000 resamples) otherwise. AP, average precision; Int., calibration intercept; Slope, calibration slope.

| Model                 | Acc.                          | Bal. acc.                     | Prec.                         | Sens.                         | Spec.                         | F1                            | MCC                           | ROC-AUC                       | Brier                         | AP           | Int.          | Slope        |
|-----------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|--------------|---------------|--------------|
| Stacking + RFECV      | <b>0.766</b><br>(0.701–0.831) | <b>0.777</b><br>(0.711–0.842) | <b>0.629</b><br>(0.516–0.736) | <b>0.815</b><br>(0.709–0.915) | <b>0.740</b><br>(0.649–0.823) | <b>0.710</b><br>(0.615–0.797) | <b>0.532</b><br>(0.403–0.658) | <b>0.822</b><br>(0.756–0.888) | <b>0.177</b><br>(0.143–0.213) | <b>0.696</b> | <b>-0.624</b> | <b>0.866</b> |
| Stacking              | <b>0.747</b><br>(0.675–0.812) | <b>0.758</b><br>(0.690–0.826) | <b>0.606</b><br>(0.493–0.712) | <b>0.796</b><br>(0.688–0.904) | <b>0.720</b><br>(0.631–0.804) | <b>0.688</b><br>(0.592–0.778) | <b>0.494</b><br>(0.362–0.625) | <b>0.819</b><br>(0.752–0.885) | <b>0.177</b><br>(0.143–0.213) | <b>0.702</b> | <b>-0.623</b> | <b>0.865</b> |
| Logistic regression   | <b>0.740</b><br>(0.675–0.805) | <b>0.740</b><br>(0.670–0.811) | <b>0.606</b><br>(0.491–0.722) | <b>0.741</b><br>(0.625–0.862) | <b>0.740</b><br>(0.649–0.825) | <b>0.667</b><br>(0.564–0.763) | <b>0.464</b><br>(0.325–0.606) | <b>0.811</b><br>(0.744–0.878) | <b>0.181</b><br>(0.147–0.219) | <b>0.683</b> | <b>-0.629</b> | <b>0.756</b> |
| LR + LR-ranked RFECV  | <b>0.734</b><br>(0.662–0.805) | <b>0.731</b><br>(0.660–0.806) | <b>0.600</b><br>(0.480–0.714) | <b>0.722</b><br>(0.608–0.846) | <b>0.740</b><br>(0.649–0.825) | <b>0.655</b><br>(0.551–0.753) | <b>0.447</b><br>(0.309–0.591) | <b>0.809</b><br>(0.741–0.876) | <b>0.182</b><br>(0.147–0.220) | <b>0.684</b> | <b>-0.634</b> | <b>0.746</b> |
| Random forest + RFECV | <b>0.734</b><br>(0.662–0.799) | <b>0.723</b><br>(0.648–0.794) | <b>0.607</b><br>(0.482–0.724) | <b>0.685</b><br>(0.564–0.804) | <b>0.760</b><br>(0.673–0.840) | <b>0.643</b><br>(0.537–0.739) | <b>0.434</b><br>(0.290–0.573) | <b>0.826</b><br>(0.761–0.892) | <b>0.166</b><br>(0.134–0.201) | <b>0.711</b> | <b>-0.338</b> | <b>0.776</b> |
| XGBoost + RFECV       | <b>0.760</b><br>(0.688–0.825) | <b>0.734</b><br>(0.660–0.805) | <b>0.660</b><br>(0.528–0.781) | <b>0.648</b><br>(0.523–0.776) | <b>0.820</b><br>(0.743–0.888) | <b>0.654</b><br>(0.543–0.752) | <b>0.470</b><br>(0.321–0.605) | <b>0.824</b><br>(0.758–0.891) | <b>0.171</b><br>(0.132–0.215) | <b>0.697</b> | <b>+0.039</b> | <b>0.559</b> |
| SVM                   | <b>0.727</b><br>(0.656–0.799) | <b>0.684</b><br>(0.609–0.759) | <b>0.630</b><br>(0.489–0.765) | <b>0.537</b><br>(0.407–0.672) | <b>0.830</b><br>(0.755–0.896) | <b>0.580</b><br>(0.457–0.692) | <b>0.383</b><br>(0.226–0.534) | <b>0.810</b><br>(0.740–0.880) | <b>0.169</b><br>(0.135–0.205) | <b>0.684</b> | <b>+0.004</b> | <b>0.844</b> |

## S5 Feature-selection frequencies

**Table S3:** Percentage of the 50 outer folds in which each candidate predictor was retained by the common random-forest-ranked RFECV and by coefficient-ranked RFECV for logistic regression. Derived features are deterministic functions of the original variables.

| Predictor                | Type     | RF-ranked (%) | LR-ranked (%) |
|--------------------------|----------|---------------|---------------|
| Pregnancies              | original | 58            | 94            |
| Glucose                  | original | 100           | 100           |
| BloodPressure            | original | 68            | 2             |
| SkinThickness            | original | 56            | 0             |
| Insulin                  | original | 46            | 26            |
| BMI                      | original | 100           | 82            |
| DiabetesPedigreeFunction | original | 94            | 92            |
| Age                      | original | 68            | 94            |
| Glucose_BMI              | derived  | 100           | 78            |
| Glucose_Age              | derived  | 100           | 84            |
| HOMA_IR_surrogate        | derived  | 100           | 72            |
| Insulin_Glucose_Ratio    | derived  | 68            | 80            |
| BMI_Age                  | derived  | 100           | 38            |
| Obese                    | derived  | 6             | 74            |
| Hyperglycemic            | derived  | 38            | 26            |
| Pedigree_Age             | derived  | 100           | 64            |

## S6 TRIPOD+AI reporting map

Table S4 maps the main TRIPOD+AI reporting domains (Collins et al., BMJ 2024;385:e078378) to the manuscript. The domain descriptions are paraphrased. Before submission, the author should complete the official 27-item checklist, available from the TRIPOD website, using this map.

## S7 Software environment and reproduction

All results were produced by repository commit e23102d of <https://github.com/subrahmanyamtanala-rgb/T2D> with Python 3.11, scikit-learn 1.9.1, XGBoost 3.2.0, LightGBM 4.7.0, SHAP, NumPy, pandas and SciPy, using random seed 42 throughout. The dataset is included in the repository as `data/diabetes.csv`; it is the 768-record PIMA Indians Diabetes Database with a header row added. To reproduce:

```
pip install -r requirements.txt shap
python -m t2d.analysis --stage cv          # ~40 min on 4 CPU cores
python -m t2d.analysis --stage holdout     # ~7 min
python -m t2d.analysis --stage missing    # ~35 min
paper/build.sh                             # all tables, figures and PDFs
```

Fold-level metrics, out-of-fold predictions, hold-out predictions and SHAP values are provided in `results/revision/`.

**Table S4:** TRIPOD+AI domains and where they are addressed (main text section or this supplement).

| Domain (paraphrased)                                        | Where addressed                                                        |
|-------------------------------------------------------------|------------------------------------------------------------------------|
| Title and structured abstract                               | Title; Abstract                                                        |
| Background, rationale and objectives                        | Sections 1–2                                                           |
| Data source, setting and participants                       | Section 3.1                                                            |
| Outcome definition and timing                               | Section 3.1 (Outcome definition and intended setting); Section 5.5     |
| Predictors, including derived features                      | Section 3.1; Section 3.3; Table 2                                      |
| Sample size                                                 | Section 3.1; Section 5.5 (limitation)                                  |
| Missing data                                                | Section 3.2; Section 3.9; Section 4.6; Section 5.4; Table 8            |
| Data preparation and leakage control                        | Section 3; Fig. 1; Fig. S1                                             |
| Model types, hyperparameters and building                   | Sections 3.4–3.5; Table 3                                              |
| Class imbalance handling                                    | Section 3.2 (balanced class weights); Section 4.4 (calibration effect) |
| Internal validation design                                  | Section 3.6; Fig. S1; Section S2                                       |
| Performance measures (discrimination, calibration, utility) | Sections 3.7–3.8; Section S2                                           |
| Fairness across subgroups                                   | Not assessed: single-sex, single-ethnicity cohort (Section 5.5)        |
| Results: participants and model specification               | Section 3.1; Section 4; Tables S1–S3                                   |
| Results: performance with uncertainty                       | Tables 4–8; Tables S1–S2                                               |
| Interpretation, limitations and usability                   | Section 5                                                              |
| Explainability                                              | Section 3.10; Section 4.7; Fig. 7                                      |
| Ethics, funding and conflicts                               | Declarations                                                           |
| Protocol, registration, patient and public involvement      | Not applicable or not done (retrospective analysis of public data)     |
| Data and code availability                                  | Data and code availability statement; Section S6                       |